

Packet Tracer – DHCP Across Multiple Subnets

Objectives

Part 1: Build a Multi-Subnet Network

Part 2: Configure Router Interfaces

Part 3: Configure Router as DHCP Server

Part 4: Configure DHCP Clients

Part 5: Verify DHCP Address Assignment

Part 6: Verify End-to-End Connectivity

Background / Scenario

You are deploying a small enterprise network consisting of two LANs connected by a router. Instead of manually assigning IP addresses, the router will act as a DHCP server and automatically provide IP addresses, subnet masks, and default gateways to clients in both subnets.

This lab focuses on:

- DHCP configuration
- Multiple subnet environments
- Router-based DHCP services
- Dynamic IP assignment
- Connectivity verification

At the end of the lab, both PCs should automatically obtain addresses and successfully communicate across subnets.

Instructions

Part 1: Build the Network

Step 1: Add Devices

Add the following devices:

- 1 Router (2911 or 1941)
- 2 Switches (2960)
- 2 PCs

Rename:

- Router0 → **R1**
 - Switch0 → **S1**
 - Switch1 → **S2**
 - PC0 → **PC-A**
 - PC1 → **PC-B**
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Step 2: Connect Devices

Use **Copper Straight-Through** cables.

Device Interface Device Interface

PC-A	Fa0	S1	Fa0/1
S1	Gi0/1	R1	Gi0/0
R1	Gi0/1	S2	Gi0/1
S2	Fa0/1	PC-B	Fa0

Wait until all links become green.

Network Addressing Plan

LAN 1

Device Address

R1 G0/0 192.168.10.1/24

Device Address

PC-A DHCP

LAN 2

Device Address

R1 G0/1 192.168.20.1/24

PC-B DHCP

Part 2: Configure Router Interfaces

On R1:

enable

configure terminal

interface g0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

exit

interface g0/1

ip address 192.168.20.1 255.255.255.0

no shutdown

exit

Verify:

show ip interface brief

Expected:

GigabitEthernet0/0 up up

GigabitEthernet0/1 up up

Part 3: Configure Router DHCP Service

Step 1: Exclude Router Addresses

```
ip dhcp excluded-address 192.168.10.1  
ip dhcp excluded-address 192.168.20.1
```

Step 2: Create DHCP Pool for LAN1

```
ip dhcp pool LAN10  
  
network 192.168.10.0 255.255.255.0  
  
default-router 192.168.10.1  
  
dns-server 8.8.8.8
```

Step 3: Create DHCP Pool for LAN2

```
ip dhcp pool LAN20  
  
network 192.168.20.0 255.255.255.0  
  
default-router 192.168.20.1  
  
dns-server 8.8.8.8
```

Verify DHCP Pools

```
show ip dhcp pool
```

You should see:

```
LAN10  
LAN20
```

Part 4: Configure DHCP Clients

Step 1: Configure PC-A

Navigate:

Desktop → IP Configuration

Select:

DHCP

Expected Result:

IP Address: 192.168.10.x

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Step 2: Configure PC-B

Navigate:

Desktop → IP Configuration

Select:

DHCP

Expected Result:

IP Address: 192.168.20.x

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

Part 5: Verify DHCP Assignment

On PC-A

Open Command Prompt:

ipconfig

Expected:

IP Address.....192.168.10.x

Subnet Mask.....255.255.255.0

Default Gateway...192.168.10.1

On PC-B

ipconfig

Expected:

IP Address.....192.168.20.x

Subnet Mask.....255.255.255.0

Default Gateway...192.168.20.1

Part 6: Verify Connectivity

Step 1: Ping Local Gateway

From PC-A:

ping 192.168.10.1

Should succeed.

From PC-B:

ping 192.168.20.1

Should succeed.

Step 2: Ping Across Subnets

From PC-A:

ping 192.168.20.2

(or whatever IP PC-B received)

Expected:

Reply from 192.168.20.x

Step 3: Ping Back

From PC-B:

ping 192.168.10.2

(or whatever IP PC-A received)

Expected:

Reply from 192.168.10.x

Troubleshooting Tips

If PC-A receives:

Default Gateway: 0.0.0.0

Check:

show run | section dhcp

Verify:

default-router 192.168.10.1

exists inside the DHCP pool.

Verify DHCP bindings:

show ip dhcp binding

Verify pool statistics:

show ip dhcp pool

Force clients to renew:

```
ipconfig /release
```

```
ipconfig /renew
```

Verify router interfaces:

```
show ip interface brief
```

What You Learned

- How DHCP automatically assigns IP addresses
 - How routers can function as DHCP servers
 - How DHCP pools are configured
 - How default gateways are distributed by DHCP
 - How multiple subnets receive addresses from different pools
 - How to verify DHCP bindings and leases
 - How routed networks allow communication between different LANs
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Challenge Extensions (Optional)

Try modifying the lab to deepen your understanding of DHCP and routed networks.

Challenge 1: Reserve Addresses for Network Devices

Challenge 2: Add a Third Subnet

Challenge 3: Configure Custom DNS Servers

Challenge 4: View DHCP Bindings

Challenge 5: Release and Renew Addresses

Challenge 6: Observe the DORA Process